ROLL NO: | | | | | | | | | | |

(To fill the roll no is compulsory)

INVERTIS UNIVERSITY BAREILLY

EVEN SEMESTER EXAMINATION - 2024-25
DIPLOMA (CSE)
Semester: IV
SUBJECT CODE: DCS401
DATA STRUCTURES WITH C

Time: 03:00 Hrs. **Maximum Marks: 60**

NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS. ATTEMPT ALL THE SECTIONS.

SECTION – A

NOTE: Attempt all parts of the following question. (12 Marks)

Q. 1. Define the following terms: (1×6 = 6 Marks)

(i) Primitive Data Structures
(ii) Time Complexity
(iii) Parent and Child of a tree
(iv) Binary Tree
(v) Hashing
(vi) Recursive Function

Q. 2. Answer the following questions: (1×6 = 6 Marks)

(i) What is the key difference between linear and non-linear data structures?
(ii) List advantages of an array.
(iii) Define big O notation.
(iv) How does a queue differ from a stack?
(v) What is the advantage of lined list?
(vi) What is the time complexity of linear search in the best and worst cases?

SECTION – B

NOTE:Attempt all questions. (6×2×2 = 24 Marks)

Q.3.(i) (a) Which functions/expressions have the best and worst running times?

OR

(b) Differentiate between recursive and iterative functions with examples.

(ii) (a) How do you measure the performance of an algorithm?

OR

(b) Convert the following infix expressions into postfix Expressions (step-wise): **(**A+B)*C+(D-E)/F+G

Q.4.(i) (a) What are the main operations of a stack?

OR

(b) What are the real-life applications of stacks and queues?

(ii)(a) What is a singly linked list?

OR

(b) Define overflow and underflow in the context of linked lists.

SECTION – C

NOTE: Attempt all questions. (12×2 = 24 Marks)

Q.5. (a) Write the algorithm for **merge sort** and explain how the divide-and-conquer technique is used. Analyze its time complexity.

OR

(b) Write and explain the bubble sort algorithm.

Q.6. (a) What are multidimensional array? Write a program using arrays to multiply matrices:

Matrix 1

$$\begin{Bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{Bmatrix}$$

Matrix 2

$$\begin{Bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{Bmatrix}$$

OR

(b) Describe linear data structures with suitable examples and applications**.**